

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Summer-2023 Mid-term Examination
Program: B.Sc. in CSE (Batch – 15th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: CSE 0533 1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x1=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Understanding

1. What is the angular momentum of the 2nd orbit of atom?
- A) h
 - B) $2h$
 - C) $2h/\pi$
 - D) h/π
 - E) $h/2\pi$

Cognitive Understanding

2. Which of the following is the thermometric properties of thermocouple thermometer?
- A) Resistance
 - B) Volume
 - C) Electromotive force
 - D) Brightness
 - E) Pressure

Cognitive Knowledge

3. What is the unit of molar specific heat?
- A) calJ^{-1}
 - B) Jcal^{-1}
 - C) JK^{-1}
 - D) KJ^{-1}
 - E) calJ

Cognitive Application

4. The area of the curve between which two quantities indicates the amount of work done? (*Where, P- Pressure, V-Volume, S-Entropy, T-temperature, W-external work done*)
- A) P & T
 - B) P & S
 - C) P & W
 - D) P & V
 - E) V & T

Cognitive Knowledge

5. How many Isothermal compression steps are present in the Carnot's cycle?
- A) 2
 - B) 3
 - C) 4
 - D) 5
 - E) 6

Cognitive Knowledge

6. The pressure of the gas generated by the collision of the
- A) atoms with each others.
 - B) molecules with each others.
 - C) molecules with the wall of the container.
 - D) external air with the wall of the container.
 - E) molecules with the external air molecules.

Cognitive Application

7. A heat engine after doing work in each cycle rejects 70% of heat absorbed from the source. What will be the efficiency (%) of the engine?
- A) 70
 - B) 66
 - C) 48
 - D) 30
 - E) 23

Cognitive Understanding

8. For Adiabatic process, which of the following is considered as constant?
- A) Temperature (T)
 - B) Volume (V)
 - C) Pressure (P)
 - D) Entropy (S)
 - E) Gibbs free energy (G)

Cognitive Application

9. When heat is supplied to a gas of volume 100m^3 at normal pressure, the volume changes to 100.2 m^3 . How much work done (J) by the gas?
- A) 40460
 - B) 30360
 - C) 20200
 - D) 10160
 - E) 0.0060

Cognitive Knowledge

10. Which will be the lower orbit number for the Paschen series of a hydrogen atom?
- A) 1
 - B) 2
 - C) 3
 - D) 4
 - E) 5

Signature of Examiner

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PART-B: SAQ

Time: 1 Hour and 20 Minutes

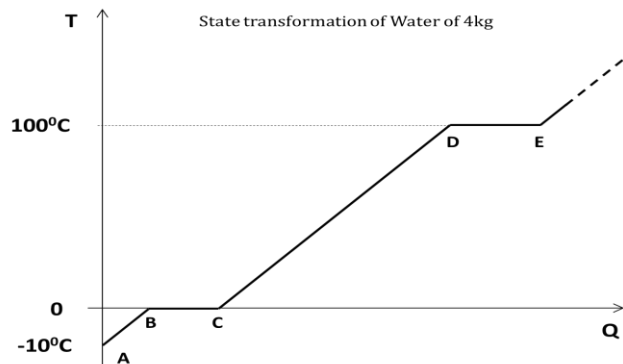
Total Marks: 4x5=20

INSTRUCTIONS: Answer question no. 01 (one) and any 03 (three) from the rest. Marks are shown by side.

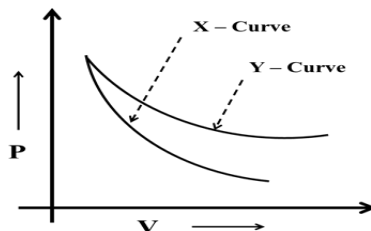
Question No.

Marks

1. See the transformation curve and answer the questions that follows:



- Specify entropy. [1]
 - Find the value of BC. [1]
 - Compare the change in entropy for CD and DE portion of the above-mentioned curve. [3]
- 2.
- State the principle of the temperature measurement. [1]
 - Which metal is mostly used in resistance thermometers and why? [2]
 - A platinum resistance thermometer reads 12Ω and 20Ω at ice point and steam point respectively. Find the temperature at which it reads 36Ω . [2]
- 3.
- Disclose a condition of elastic collision. [1]
 - Differentiate real and ideal gases graphically and justify your answer. [1]
 - Construct the Van-der-wall equation in the correction of real gas equation. [3]
4. See the figure for thermodynamic system and answer the questions that follows:



- Specify the X – Process. [1]
 - Identify which curve follows the Boyle's law. [1]
 - Is it possible to establish a relation of slope of above two curves? – Justify your answer. [3]
- 5.
- Indicates the significance of positive and negative Gibbs free energy. [1]
 - Find the wave number of the Balmer series of hydrogen atom. [2]
 - Calculate the energy of an electron of 2nd orbit of Helium atom. [2]